

An aerial photograph of the University of British Columbia Okanagan campus. The image shows several large, modern brick buildings with flat roofs, some featuring solar panels. The campus is surrounded by lush green trees and lawns. In the background, there are rolling hills and mountains under a clear blue sky with a few wispy clouds. The lighting suggests it's either early morning or late afternoon, with long shadows cast across the landscape.

Special Topics: Applied Computer Vision and Deep Learning

UBCO Master of Data Science – DATA 589



Topics

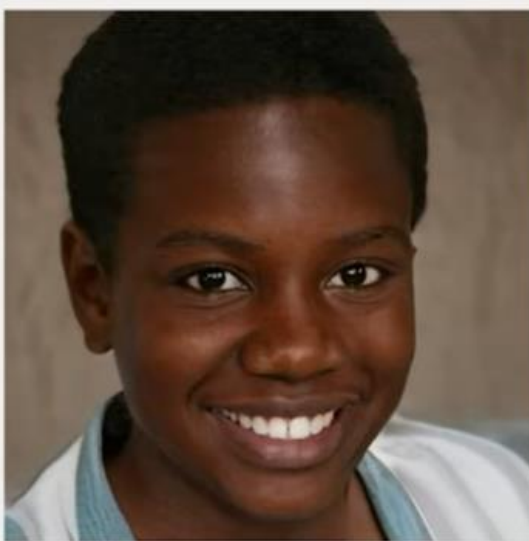
Generative Models

- Autoencoders
- Variational Auto Encoders (VAEs)
- Generative Adversarial Networks - GANs
- Diffusion Models

Which face is real?



A



B



C

Supervised vs Unsupervised Learning

Supervised Learning

- **Data:** (x, y)
 x : input data, y : label
- **Goal:** Learn mapping $f(x) \rightarrow y$
- **Examples:** Classification, Regression, Object Detection, Segmentation

Unsupervised Learning

- **Data:** x (no labels)
- **Goal:** Learn hidden structure of data
- **Examples:** Clustering, Dimensionality Reduction, Density Estimation

What if we want to generate new data that looks like x ?

This naturally leads us to **Generative Models**, where instead of predicting labels, we model the data distribution itself.

“Generative models try to learn $p(x)$ —the distribution of the data.”

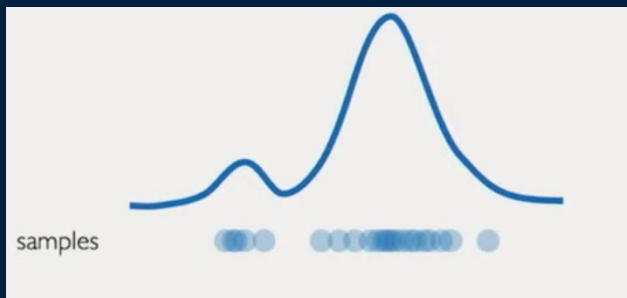
Generative Modeling

Goal: Given training samples from a data distribution, learn a model that approximates that distribution.

Density Estimation

Learn a probability distribution:

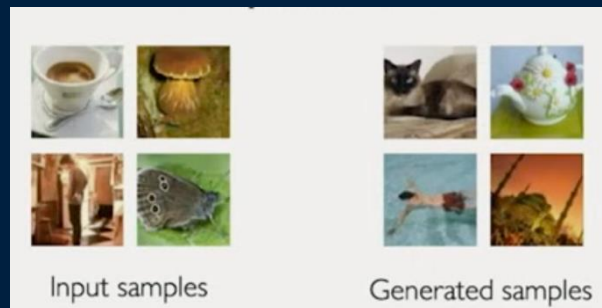
$$P_{model}(x) \approx P_{data}(x)$$



Sample Generation

If we can model $P_{model}(x)$, we can generate new samples:

$$x_{new} \sim P_{model}(x)$$



Why Generative Models? (Debiasing)

Generative models can uncover hidden structure and variation in data.

If a dataset is biased — for example, mostly similar skin tones, poses, or lighting — the model will learn that bias.

By learning the full data distribution, generative models help us:

- Detect imbalance in datasets
- Identify missing variations
- Generate diverse synthetic samples
- Improve fairness and representation
- This allows us to build more robust and less biased AI systems.

Underlying features in a dataset



Homogeneous skin color, pose

VS



Diverse skin color, pose, illumination

How can we use this information to create fair and representative dataset?

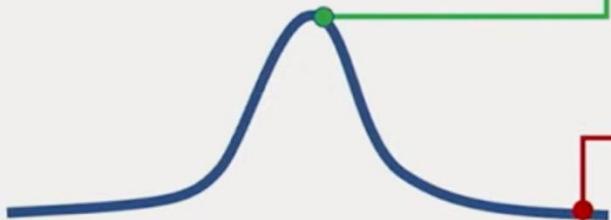
Outlier Detection

- **Problem:** How can we detect when we encounter something new or rare?
- **Strategy:** Leverage generative models, detect outliers in the distribution
- Use outliers during training to improve even more!

95% of Driving Data:
(1) sunny, (2) highway, (3) straight road



Detect outliers to avoid unpredictable behavior when training



Edge Cases



Harsh Weather



Pedestrians

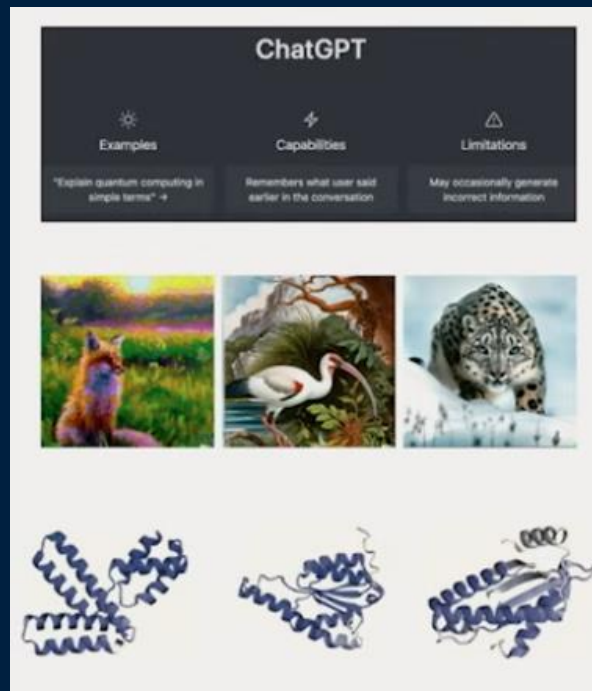
Why Generative Models?

Generative models learn a probability distribution $P(x)$ over data.

Once we approximate this distribution, we can **sample from it** to create new data instances:

$$x_{new} \sim P_{model}(x)$$

This is fundamentally different from supervised learning: we are not predicting labels, we are generating data itself.



Question

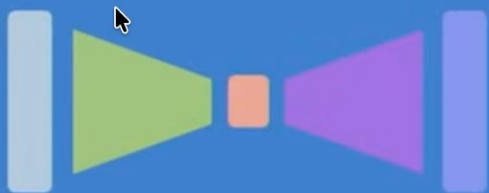
Question: What is the main goal of a generative model?

- A. To predict labels from input data
- B. To compress data into smaller vectors only
- C. To learn the underlying data distribution and generate new samples
- D. To remove the need for neural networks
- E. To increase the number of labels in a dataset

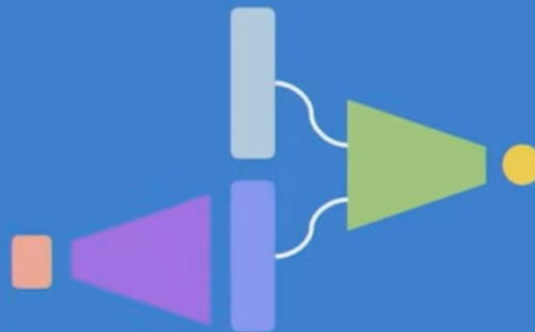
Latent Variable Models

Latent variable models can be broken down to autoencoders and GANs.

Autoencoders and Variational Autoencoders (VAEs)



Generative Adversarial Networks (GANs)



What Is a Latent Variable?

Observable variables: directly measured data x

Latent variables: hidden factors z that generate the data

The visible tip above water represents:

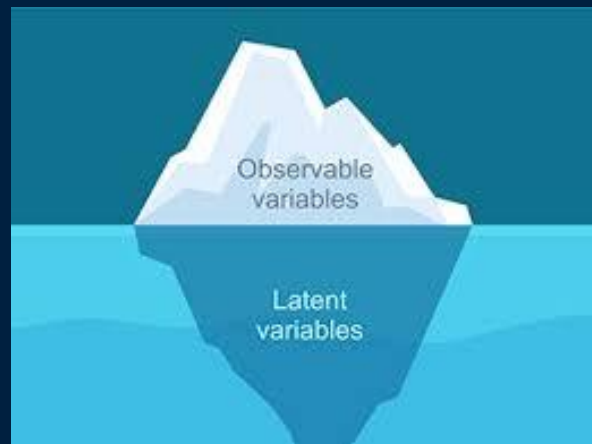
- Pixels in an image
- Words in a sentence
- Audio waveform
- Protein structure

These are observable variables x .

But most of what determines them lies underneath — the latent variables z .

For images, z might represent:

- Identity
- Pose
- Illumination
- Expression
- Style



Autoencoders

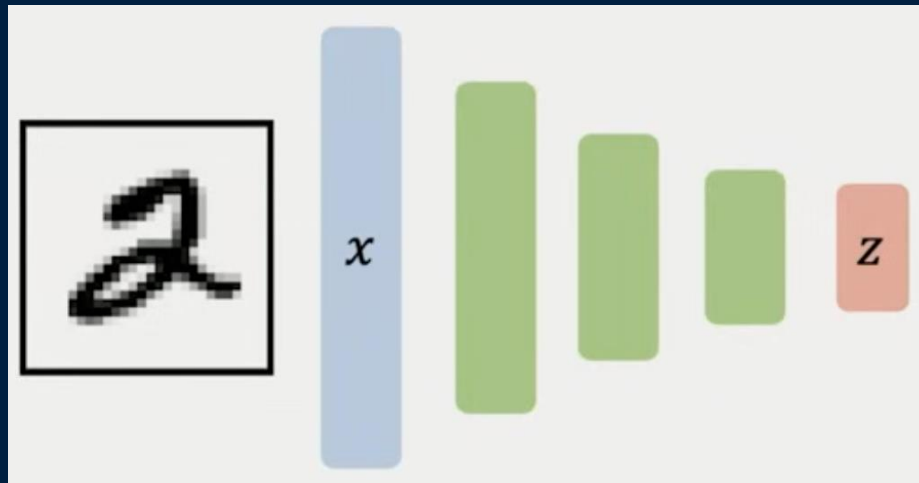
Autoencoders: Unsupervised method for learning a **low-dimensional representation** from unlabeled data.

Autoencoders are one of the simplest latent variable models.

They answer this question:

Can we compress data into a smaller representation and then reconstruct it back?

Encoder learns mapping from data x to a low dimensional latent space, z



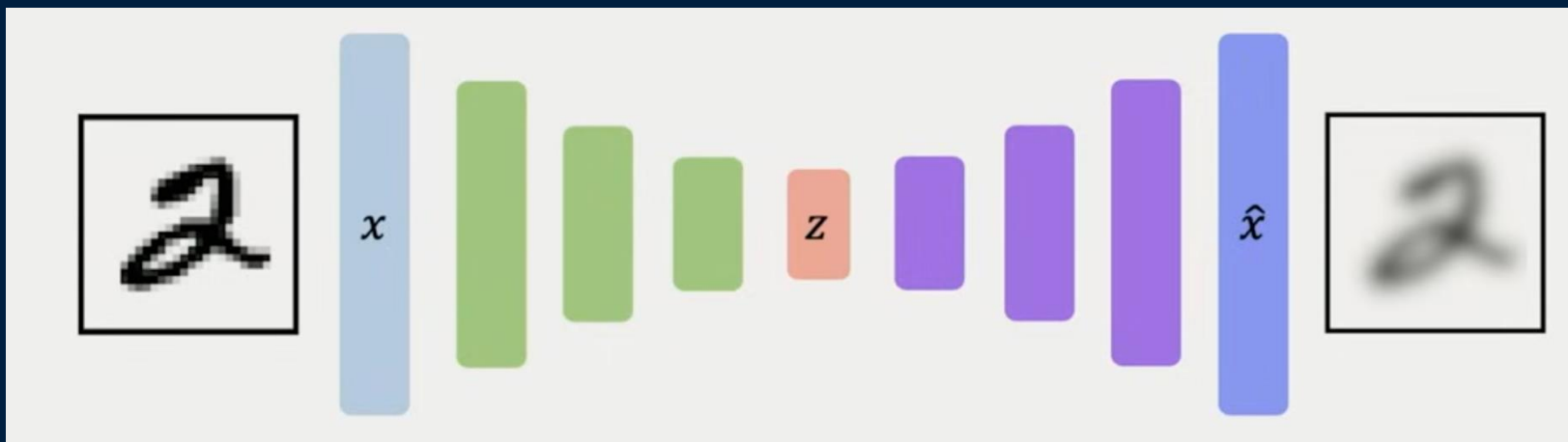
Autoencoders

How can we learn this latent space? Train the model to use these features to reconstruct the original data.

An autoencoder has two parts:

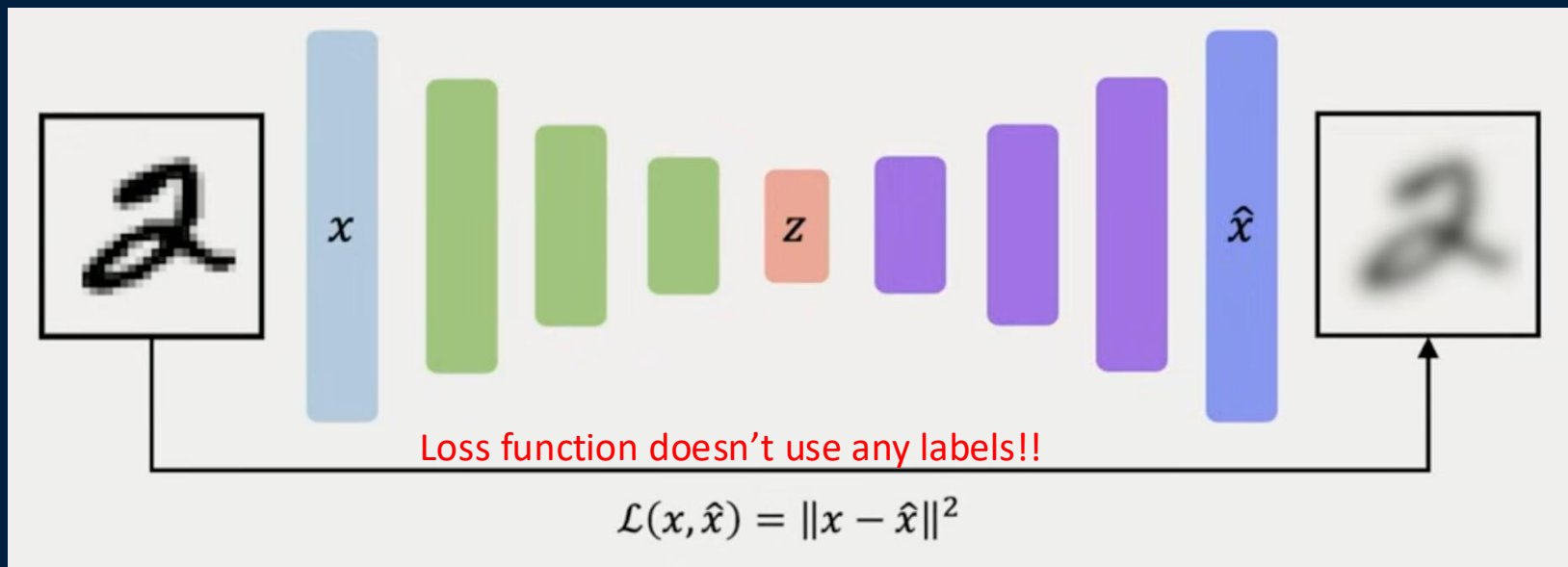
Encoder: Maps high-dimensional input x to lower-dimensional latent space z .

Decoder: Maps z back to reconstructed output $\hat{x}$.



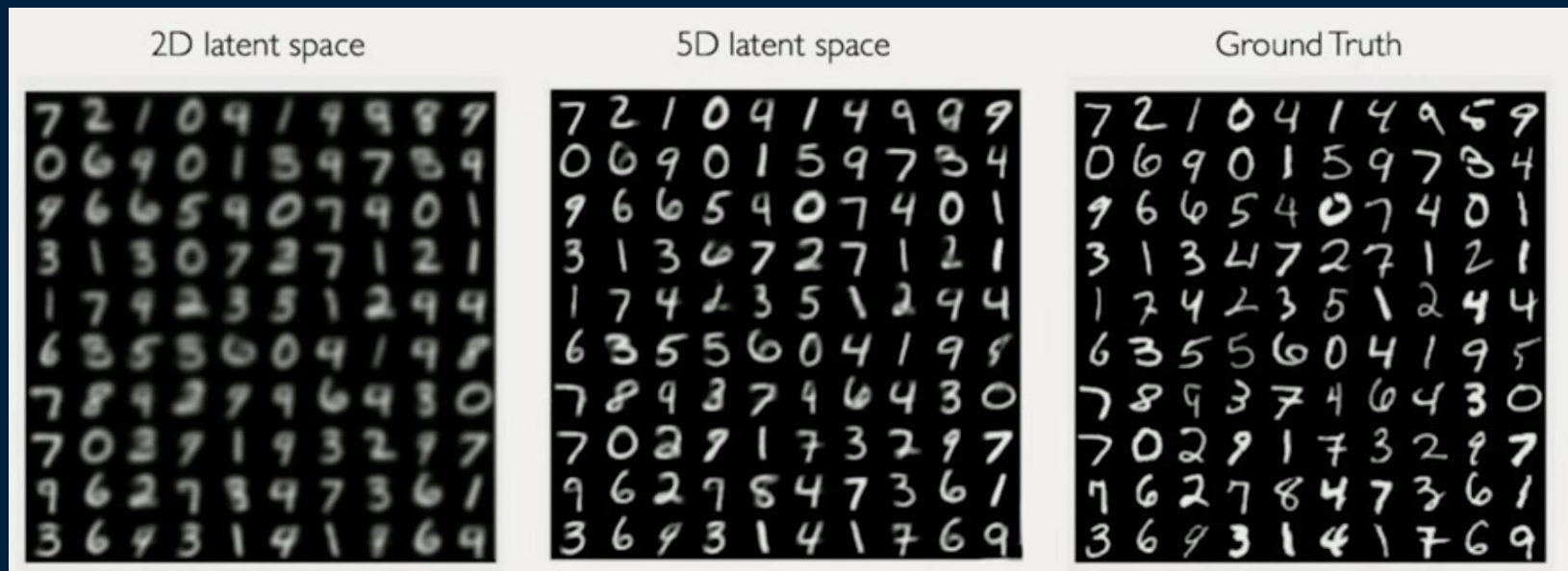
Autoencoders

How can we learn this latent space? Train the model to use these features to reconstruct the original data and **minimizing reconstruction error**. This is **Unsupervised Learning**.



Dimensionality of latent space = quality of reconstruction

- Autoencoding is a form of compression!
- Smaller latent space will force a larger training bottleneck
- The size of the latent space determines what information the model is allowed to remember.



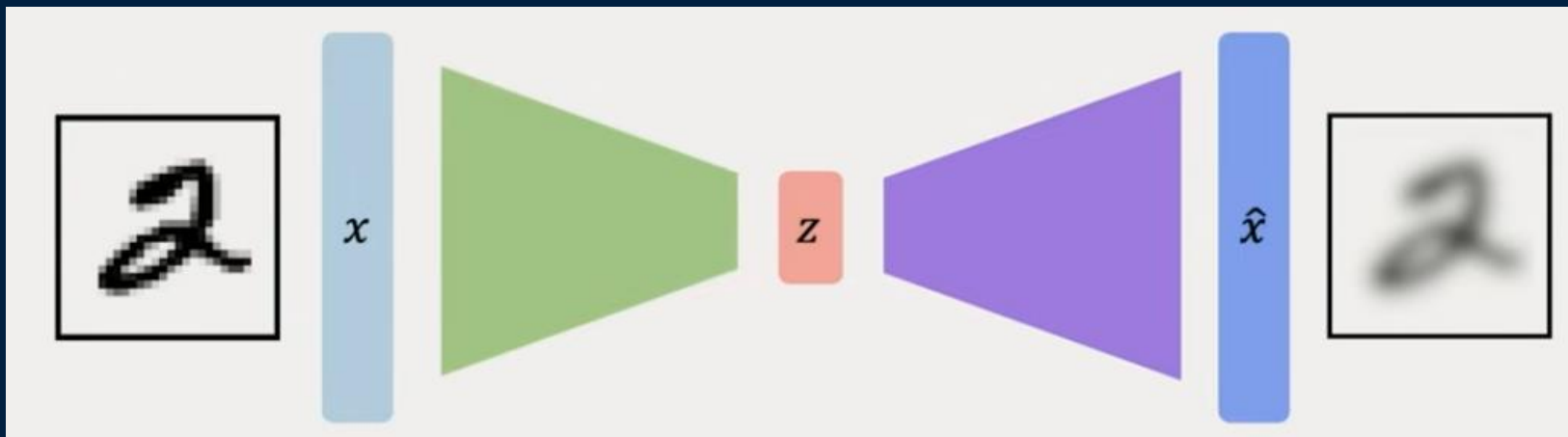
Question

Question: **Why is the bottleneck important in an autoencoder?**

- A. It makes the model train faster by increasing parameters
- B. It forces the model to compress information and learn meaningful features
- C. It guarantees that the model becomes probabilistic
- D. It replaces the decoder during inference
- E. It allows the model to use labeled data more efficiently

Variational Autoencoders (VAEs)

Autoencoders

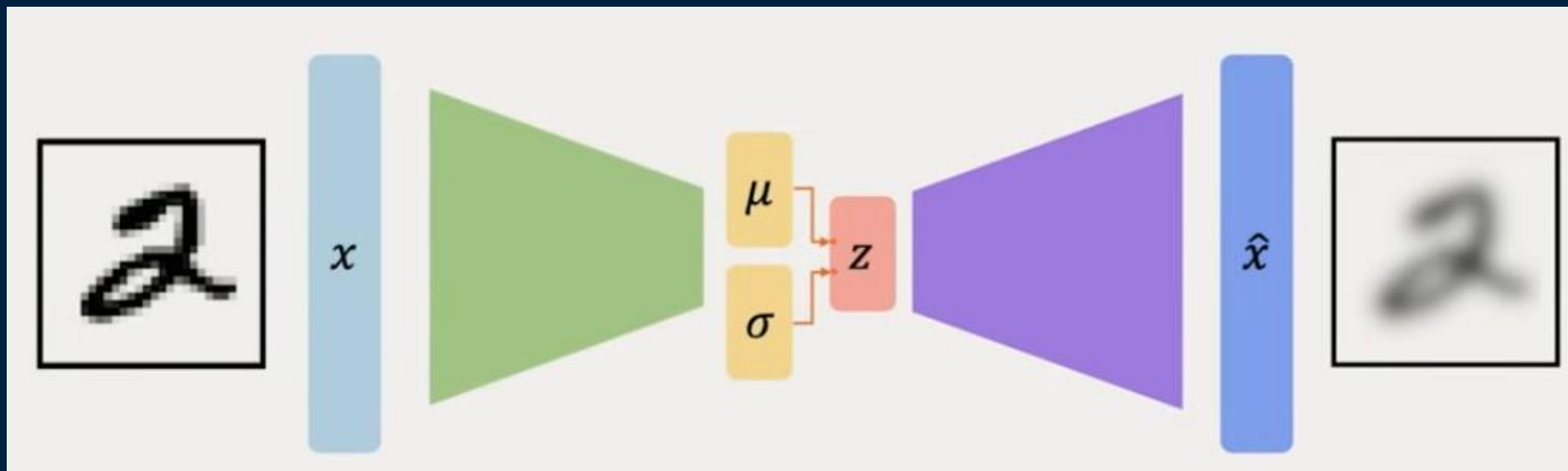


If we don't change the weights of the encoder and decoder then given an input it will generate the same output back.

This is a deterministic decoding that allows us to reconstruct an input.

We need to add some randomness and stochasticity to make a desired output.

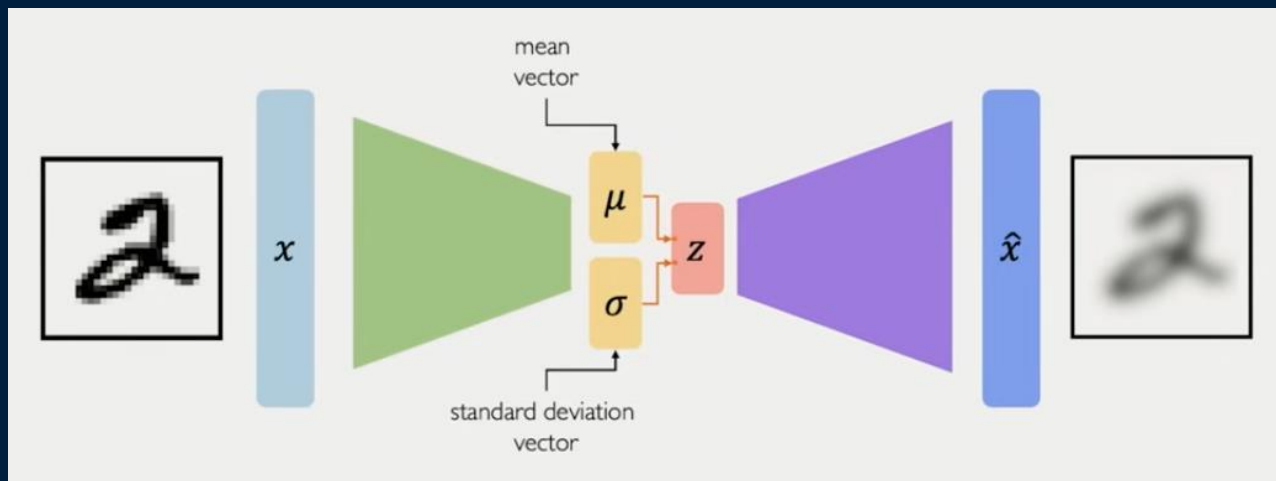
VAEs: Key Difference With Traditional Autoencoders



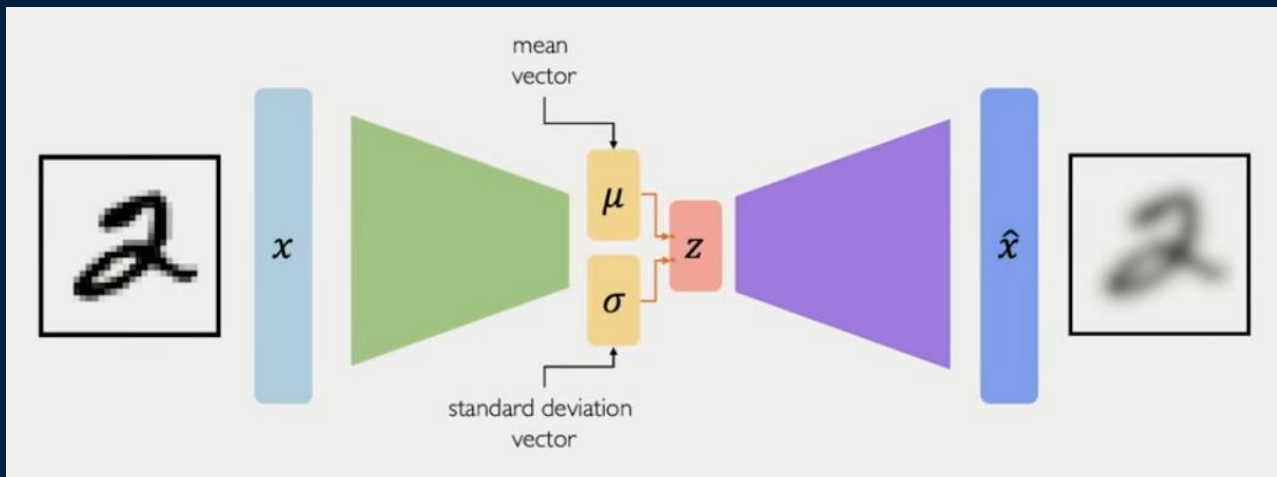
VAEs: Key Difference With Traditional Autoencoders

A Variational Autoencoder (VAE): Maps x to a **distribution over latent variables**.

A VAE doesn't learn a point in latent space — it learns a distribution.



VAEs: Key Difference With Traditional Autoencoders



Instead of producing one fixed latent code, the encoder outputs:

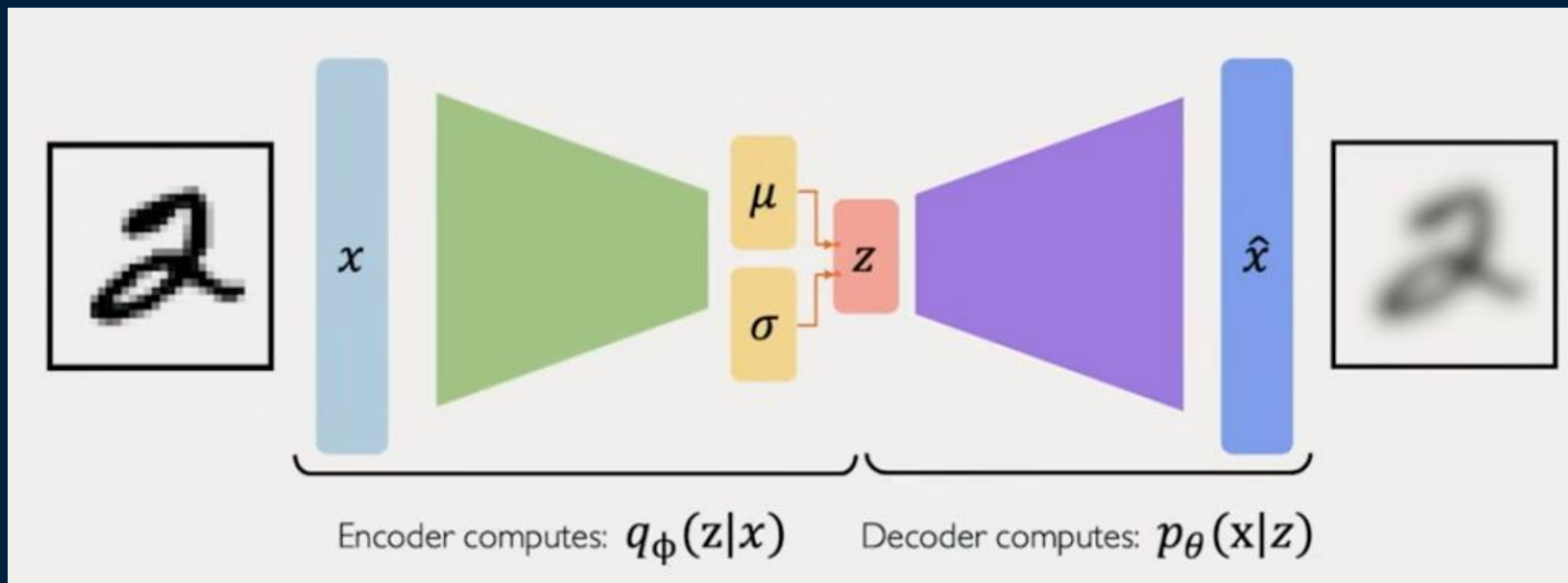
Mean vector μ

Standard deviation vector σ

We then **sample**: $z \sim \mathcal{N}(\mu, \sigma^2)$

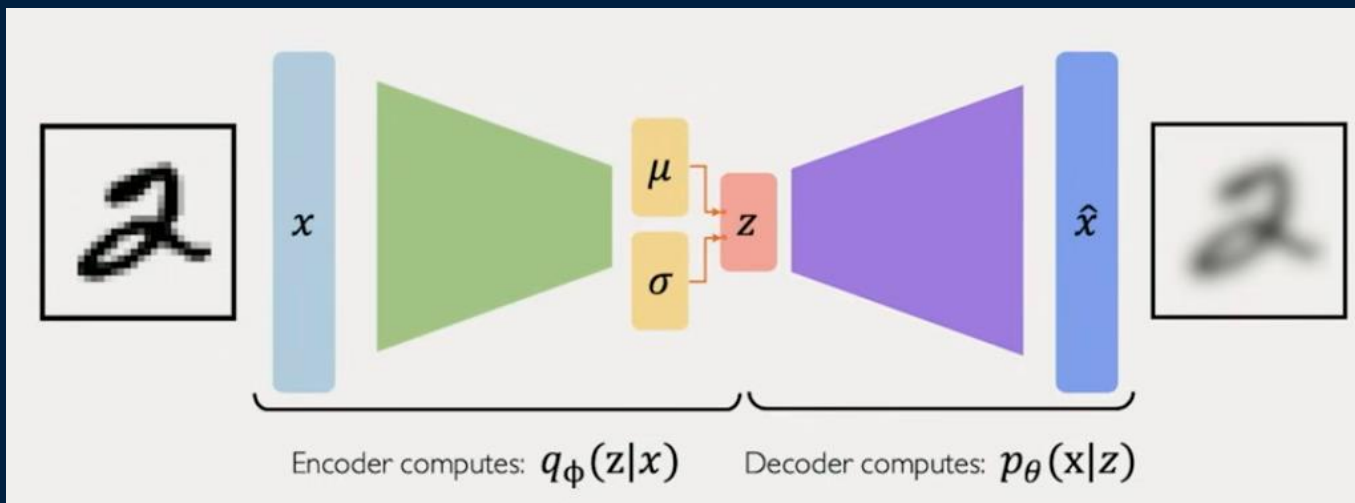
This makes the model probabilistic.

VAE Optimization



The **encoder** learns the **probability distribution** of the the **laten space** given the input data x . And the **decoder** does the reverse. Trying to **compute probability distribution of the data given the latent variable Z** .

VAE Objective Function

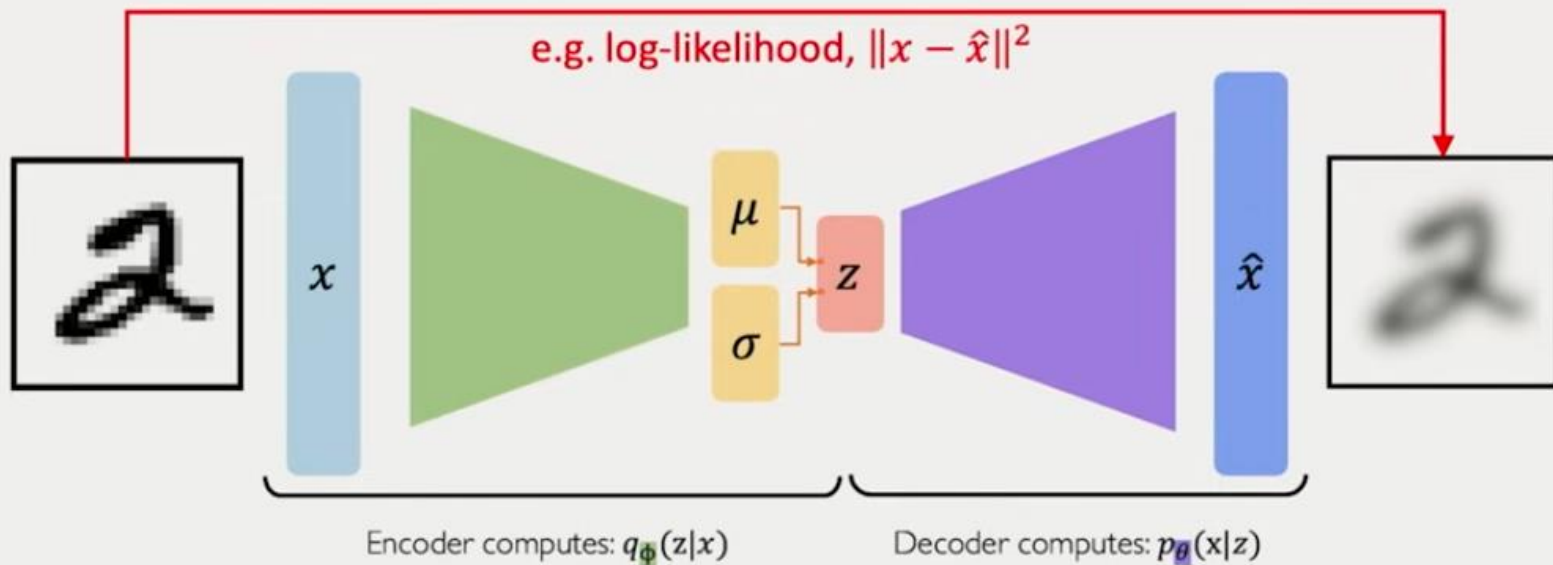


How to train it? We have two separate sets of weights ϕ and θ but we can train it end to end with one loss function:

VAE Loss

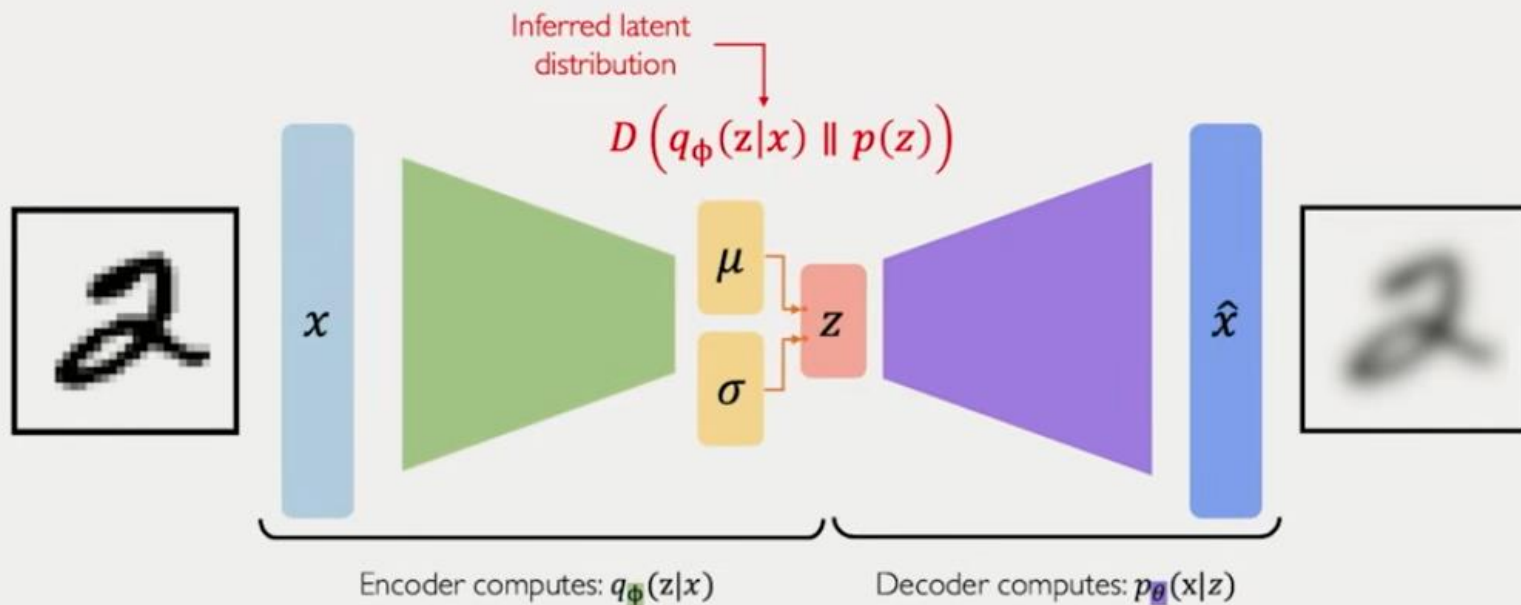
$$\mathcal{L}(\phi, \theta, x) = \text{Reconstruction term} + \text{Regularization term}$$

VAE Objective Function



$$\mathcal{L}(\phi, \theta, x) = \boxed{\text{(reconstruction loss)}} + \text{(regularization term)}$$

VAE Objective Function



$$\mathcal{L}(\phi, \theta, x) = (\text{reconstruction loss}) + (\text{regularization term})$$

VAE: Latent Perturbation

Different dimensions of z encode different interpretable latent features.

Vary one latent dimension z_i while keeping others fixed.

$$z = (z_1, z_2, \dots, z_k)$$

Modify only one coordinate and keep all other variables fixed then we see smooth transitions in generated images.



Why Latent Variable Models? (Debiasing)



Homogeneous skin color, pose

VS



Diverse skin color, pose, illumination

Different latent variables can capture different interpretable data structures, and we can use those latent variables to uncover hidden factors in our data like bias or under representative.

VAE Summary

1. Learn compressed latent representations

Encode high-dimensional data into structured latent space z .

2. Enable unsupervised learning

Train using reconstruction — no labels required.

3. Train end-to-end

4. Interpret latent factors

Explore and manipulate semantic features through latent perturbation.

5. Generate new data

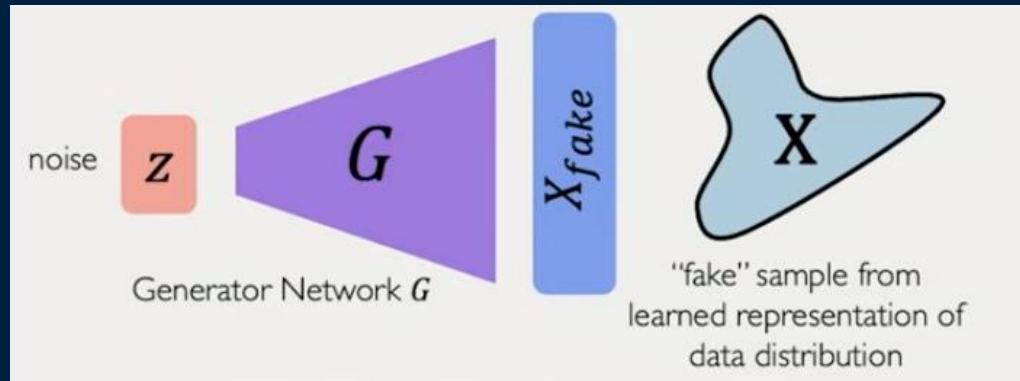
Generative Adversarial Networks (GANs)

What If We Just Want to Sample?

Instead of explicitly modeling the full data density $p(x)$,
what if we only care about generating realistic samples?

We don't model density explicitly.

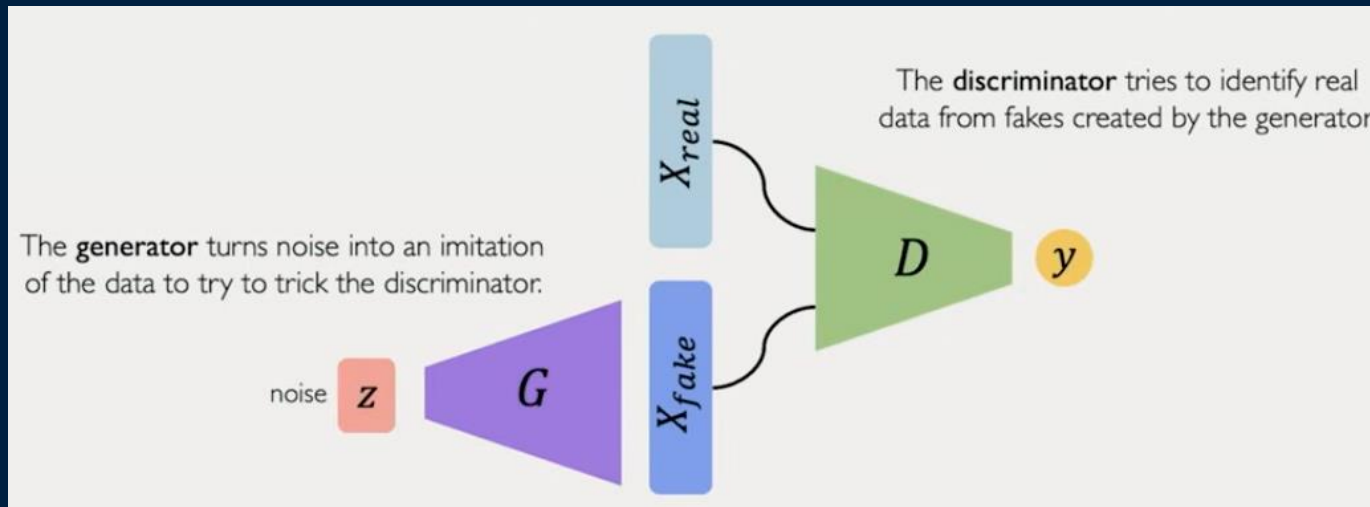
GAN learns a function that transforms simple randomness into realistic samples.



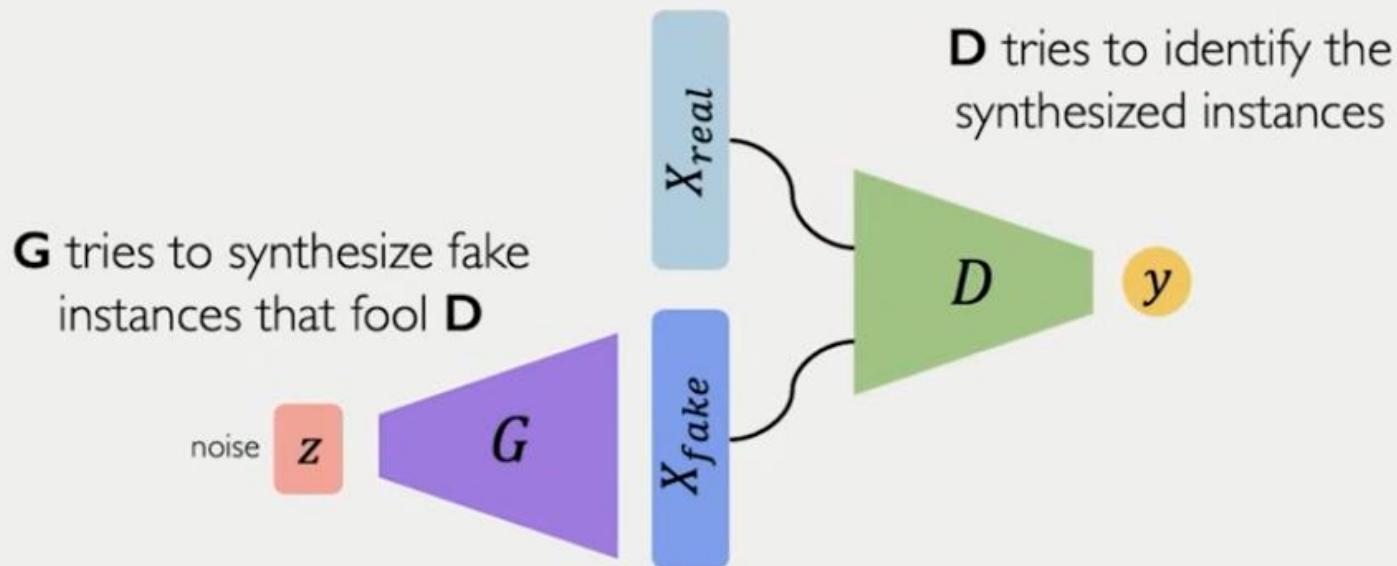
GAN Architecture

GAN can generate two data by having two neural network: a **Generator** and a **Discriminator**

The generator tries to minimize detection. The discriminator tries to maximize detection accuracy. They compete in a minimax game.

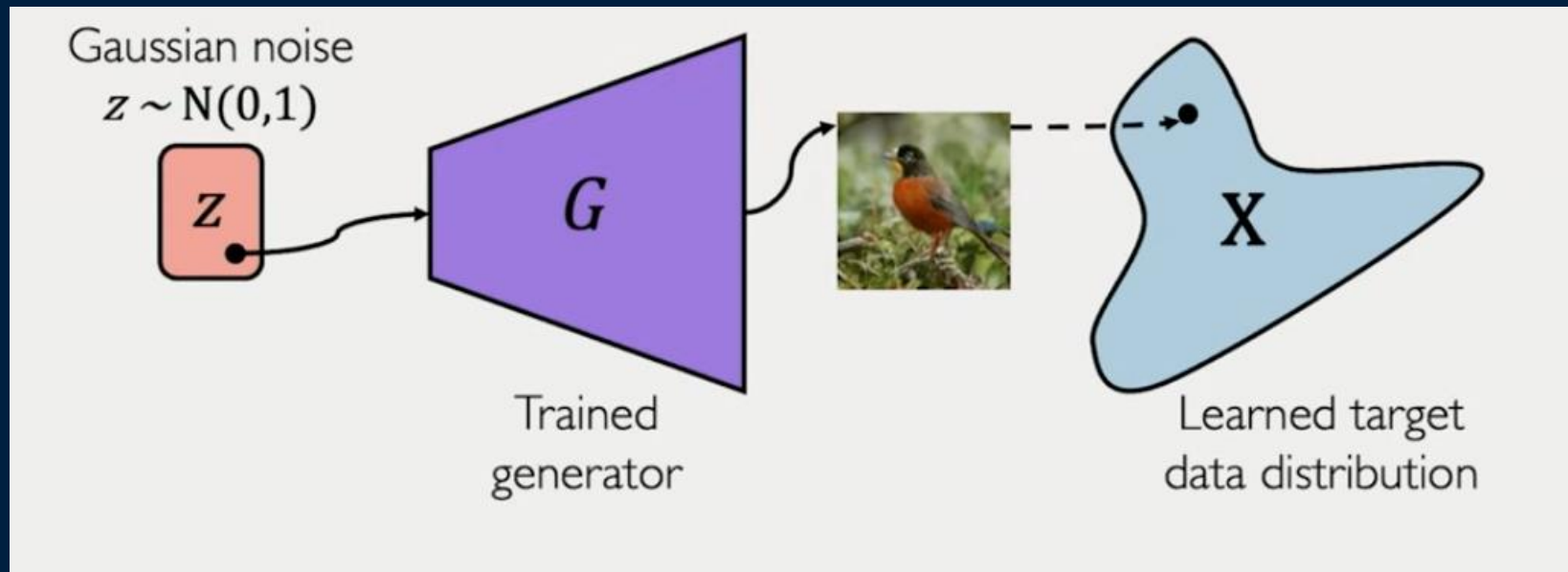


Training GAN



Training: adversarial objectives for D and G
Global optimum: G reproduces the true data distribution

GANs as Distribution Transformers



Input space: simple Gaussian Noise

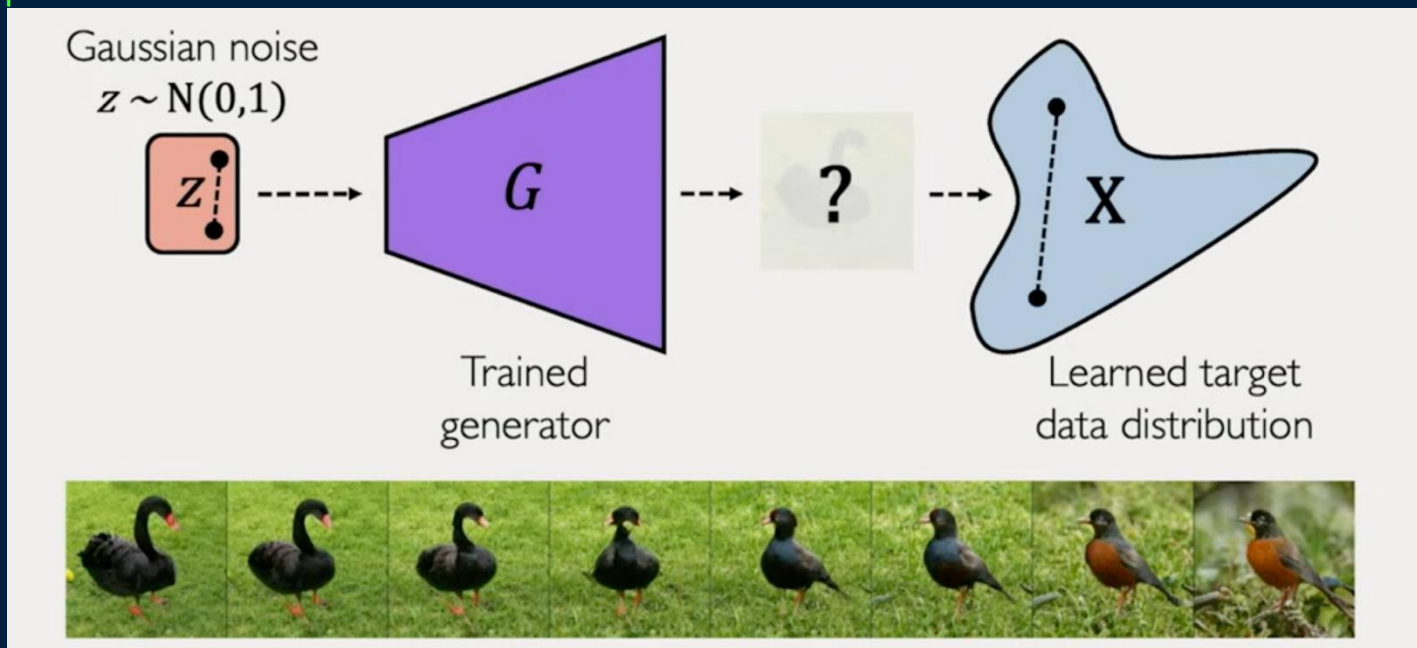
Output space: complex structured manifold (images)

The generator bends and stretches the noise space until it matches the geometry of the real data distribution.

How Does the Generator Learn the Transformation?

The generator gradually reshapes the Gaussian noise until its transformed distribution overlaps with the real data manifold.

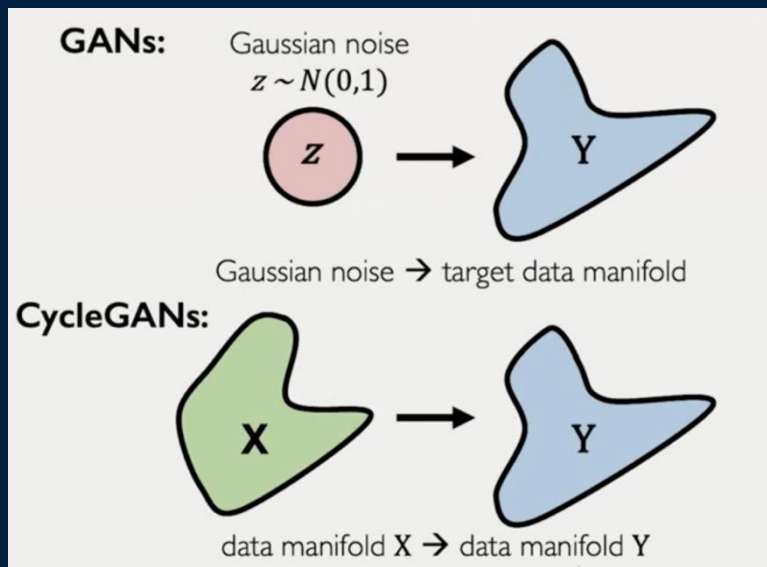
If the latent space is well learned, moving smoothly in latent space should produce smooth semantic changes in image space.



Distribution Transformations

CycleGANs: Image-to-image translation which can be used as style transformation with unpaired data!

We can also use this for audio transformation by changing audio to image spectrum (speech spectrogram) and then change style.



CycleGAN

CycleGAN learns transformations across domains with unpaired data. This is style transfer.

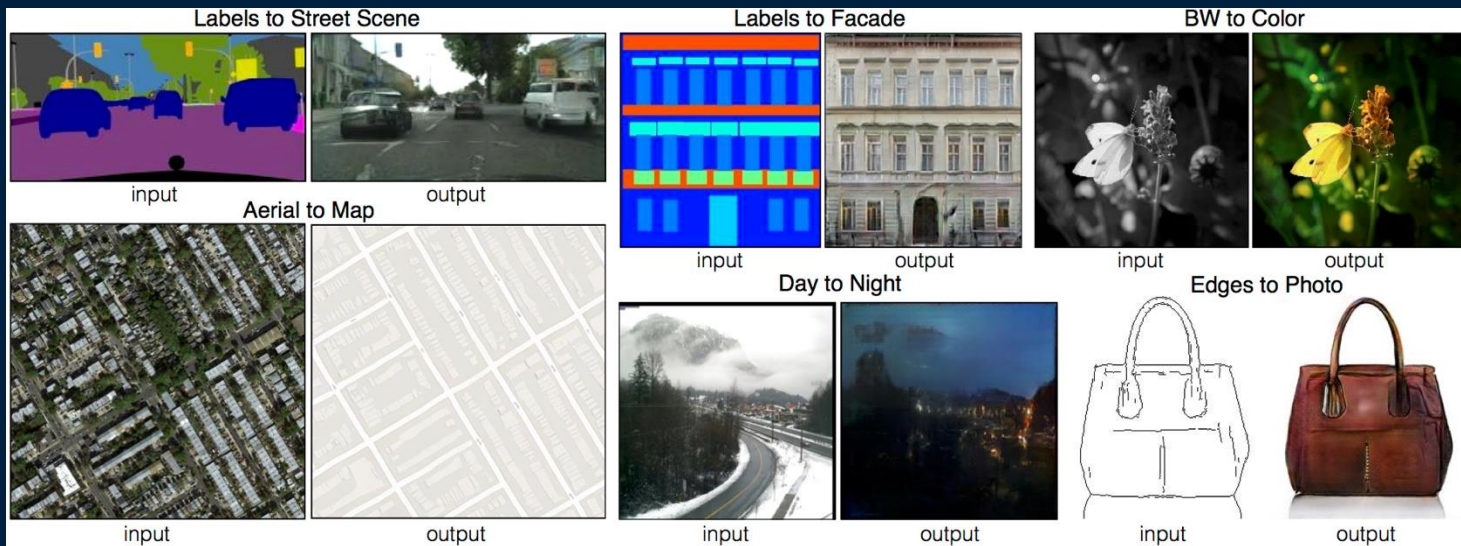
<https://github.com/junyanz/CycleGAN>



Pix2Pix - Conditional Image-to-Image Translation

Pix2pix is a paired image translation.

<https://github.com/junyanz/pytorch-CycleGAN-and-pix2pix?tab=readme-ov-file>



Question

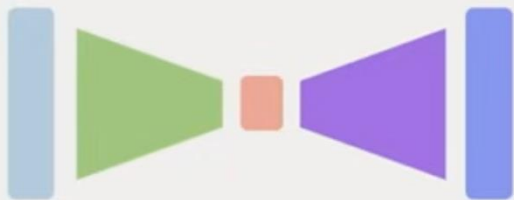
Question: **What is the main idea behind GAN training?**

- A. The encoder compresses the image into a latent vector
- B. The model learns by minimizing only reconstruction loss
- C. Noise is gradually added and then removed step by step
- D. The model learns by clustering similar images together
- E. A generator and discriminator compete in an adversarial game

Deep Generative Modeling: Recap

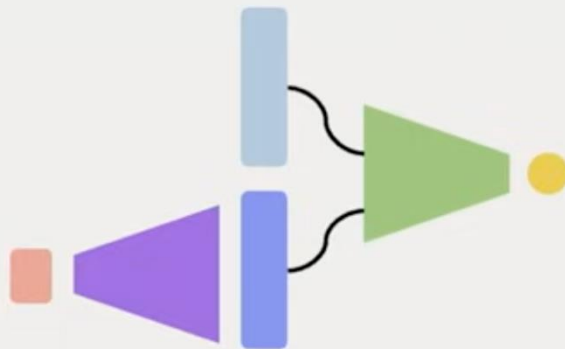
Autoencoders and Variational Autoencoders (VAEs)

Learn lower-dimensional **latent space** and **sample** to generate input reconstructions



Generative Adversarial Networks (GANs)

Competing **generator** and **discriminator** networks



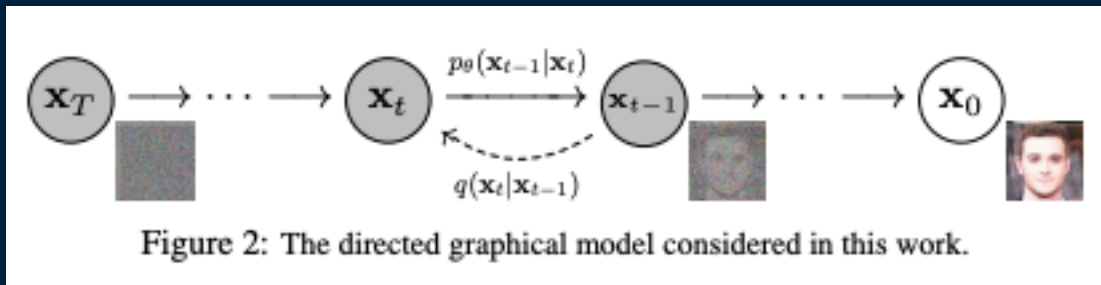
Denoising Diffusion Probabilistic Models

What Are Diffusion Models?

Generate data by learning to **reverse noise**.

Core idea:

- Start from a real image
- Gradually add noise
- Train a model to remove that noise
- Generate a new image by reversing the process



Diffusion

Compared to earlier models:

- **Autoencoder:** compresses and reconstructs
- **VAE:** learns a probabilistic latent space
- **GAN:** generates through adversarial training
- **Diffusion:** generates through iterative denoising

Why they matter:

- Stable training
- High sample quality
- Strong diversity
- Basis of modern text-to-image systems

How Diffusion Works

Two Processes

1. Forward process

$$x_0 \rightarrow x_1 \rightarrow x_2 \rightarrow \cdots \rightarrow x_T$$

Gradually add Gaussian noise until the image becomes almost pure noise.

2. Reverse process

$$x_T \rightarrow x_{T-1} \rightarrow \cdots \rightarrow x_1 \rightarrow x_0$$

Learn a neural network to remove noise step by step.

Training target:

Predict the added noise, or predict the clean image.

Training and Sampling in Practice

Training

- Sample a real image x_0
- Add noise at random time step t
- Train model to predict the noise or clean signal

Sampling

- Start from random noise
- Repeat denoising for many steps
- Final result becomes a realistic image

Practical trade-off

Strength: high quality, good diversity, stable training

Limitation: slower inference than GANs

Conditional Diffusion

Generate images with guidance.

Examples of conditions:

- Text prompt
- Class label
- Edge map
- Depth map
- Pose skeleton
- Segmentation mask
- Input image for inpainting

Why this is powerful

- Better control over outputs
- Useful for editing, design, and CV tasks
- Turns generation into a guided process

Useful Diffusion Variants Today

DDPM

- Standard diffusion formulation
- High quality, many denoising steps

DDIM

- Faster sampling
- Fewer steps, often similar quality

Latent Diffusion

- Diffusion in compressed latent space
- Much cheaper than pixel-space diffusion
- Used in Stable Diffusion / SDXL

ControlNet

- Adds spatial control
- Edges, depth, pose, segmentation

Fast modern variants

- Consistency / Latent Consistency
- Rectified Flow / Flow Matching
- Goal: fewer steps, faster generation

Question

Question: What is the key idea behind diffusion models?

- A. They generate images by competing with a discriminator
- B. They compress images into a deterministic latent code
- C. They learn to classify noisy images into categories
- D. They learn to reverse a gradual noising process step by step
- E. They generate outputs by clustering the training data

Objectives

By the end of this lecture, students should be able to:

- Explain how VAEs introduce probabilistic latent spaces and support generation
- Explain the goal of generative modeling and how it differs from supervised learning
- Describe how autoencoders learn compressed latent representations from unlabeled data
- Describe how GANs generate data through adversarial training
- Summarize how diffusion models generate data by reversing a noising process
- Compare AE, VAE, GAN, and diffusion models at a practical level



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